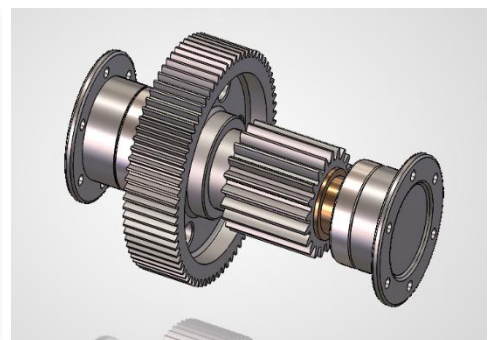
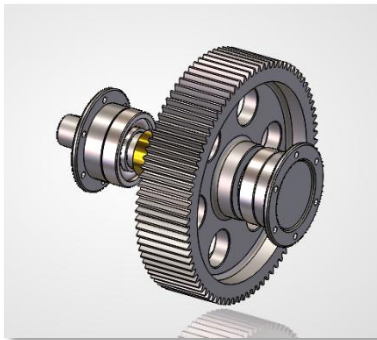
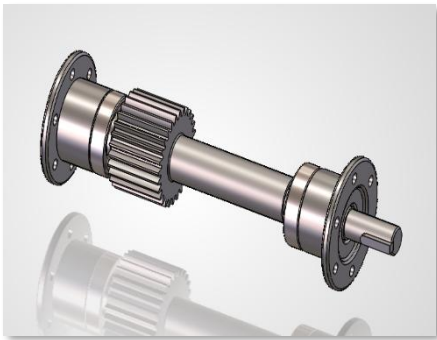
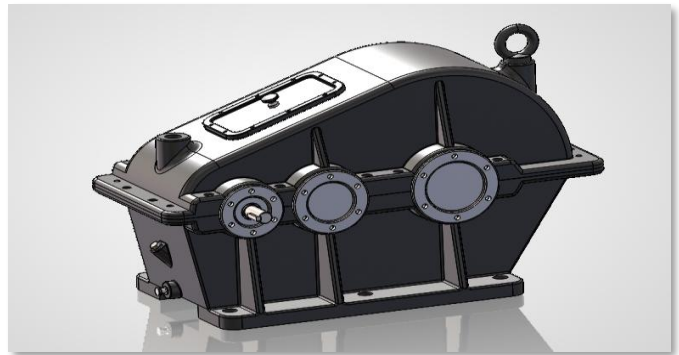
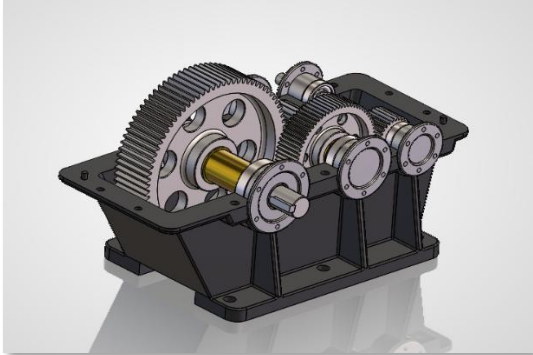


Project Title

Industrial Spur Gearbox Assembly Design and Detailing Using SolidWorks.

Project Overview

Designed and developed a complete **Spur Gearbox Assembly** in SolidWorks, including 3D part modelling, assembly creation, exploded view generation, engineering drawings, and manufacturing-ready documentation. The gearbox was created for power transmission applications requiring reliable speed reduction and torque multiplication.

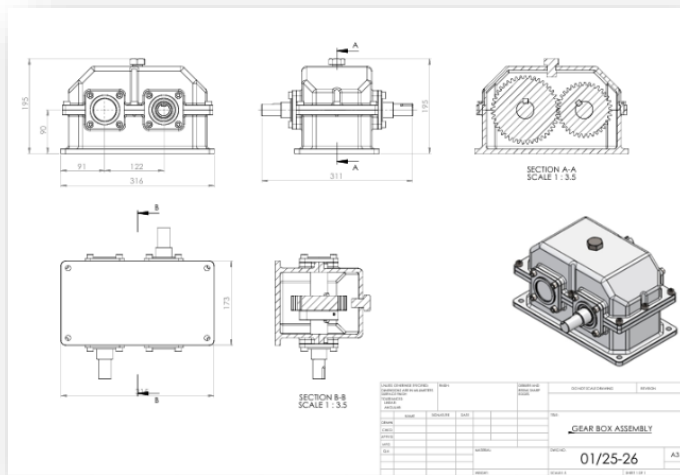


Project Objectives

- Design a robust gearbox housing assembly.
- Develop gear and shaft arrangements for power transmission.
- Create detailed manufacturing drawings.
- Generate exploded assembly views and BOM structure.
- Apply GD&T and engineering drafting standards.
- Prepare assembly documentation for production.

Software Used

- SolidWorks 2026
- AutoCAD
- Microsoft Excel (BOM Documentation)



Key Components Designed

Housing Components

- Upper Housing Cover
- Lower Housing Base
- Blunking Plug

Power Transmission Components

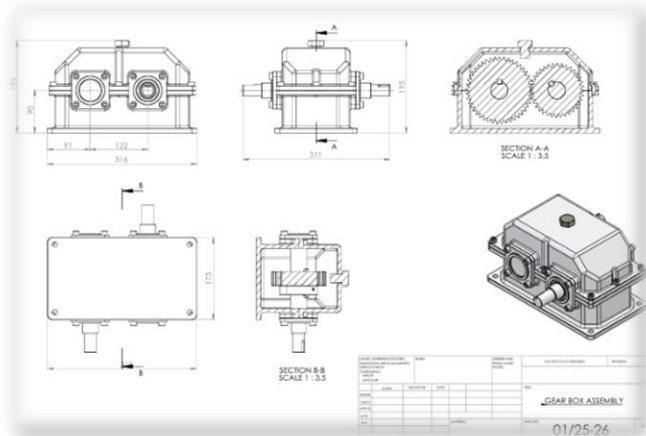
- Input Shaft
- Output Shaft
- Spur Gears
- Keys

Support Components

- Bearing Cover
- Bearings
- Rings
- Eye Bolt

Manufacturing Drawing & BOM

Item No	Part Name	Material	Qty	Remarks
1	Upper Shell Cover	GCI FG260	1	Machined
2	Base Shell Body	GCI FG260	1	Machined
3	Input Gear With 72T	EN19	1	Turned & Ground
4	1st Reduction Wheel 21T	Carbon Steel	1	Turned & Ground
5	2nd Reduction Wheel 84T	Carbon Steel	1	Hobbed
6	Input Shaft	EN 24	1	Hobbed
7	Middle Shaft	EN24	1	Purchased
8	Output Shaft	EN 24	2	Machined
9	Bearing ISO	SAE 52100	6	Standard
10	Bearing Cover	Carbon Steel	6	Standard
11	Connecting Ring	Carbon Steel	12	Standard
12	Blunking Plug	Carbon Steel	1	Standard
13	Key	Carbon Steel	5	Cut Sheet
14	Oil Stopper Plug	Carbon Steel	1	Standard
15	Eye Bolt	Carbon Steel	1	Standard
16	Fastener	Alloy Steel	60	Standard



Project Outcome

Successfully developed a complete gearbox assembly with:

- 100% parametric 3D model.
- Detailed engineering drawings.
- Exploded assembly documentation.
- Manufacturing-ready design package.
- Professional CAD documentation standards.